Smart Beta Equity Portfolio Optimization

A Rolling-Window Framework Combining Factor-Based Return Prediction with Long-Only Mean–Variance Optimization

Final Project Documentation

IDS / Math 586 — Data Science and Decision Optimization in Banking and Finance

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1. Executive Summary

This project develops a complete smart-beta equity portfolio construction pipeline that integrates statistical estimation, machine-learning-based prediction, and decision optimization in a unified rolling-window backtesting framework. The decision problem the pipeline addresses is the following: at each monthly rebalancing date, given only information available through the current month, how should an investor select a small set of attractive stocks and allocate capital across them so as to maximize realized risk-adjusted return in the following month?

Our data pipeline merges the CRSP monthly stock file with the Fama–French five-factor dataset. After standardizing dates, scaling factor returns, cleaning negative-coded prices, and applying investability filters (price $\geq \$5$, positive trading volume, at least 36 months of valid history), we construct a clean firm-month panel and a forward-looking target equal to next-month excess return. A liquidity-based candidate pool (top-100 by dollar volume each month) feeds the modeling stage. Beyond the five Fama–French factors, the feature vector now includes lagged own-stock returns at horizons of one, two, and three months, a log-market-cap size variable, *squared lagged returns* that capture non-linear momentum/reversal, and a set of *factor–size interactions* that allow factor sensitivities to vary with firm size.

Two estimation models are trained on a 36-month rolling window: an **Enhanced Ridge-regularized OLS** (RidgeCV with feature engineering) that serves as the traditional statistical baseline, and a **Random Forest** regressor designed to capture non-linear feature interactions. The OLS specification was substantially upgraded after instructor feedback that a plain linear model is unlikely to capture the cross-sectional plus time-series (CSTS) panel structure of stock returns; the enhanced specification adds non-linear terms, interaction terms, feature standardization, and cross-validated ℓ_2 regularization, producing a much fairer apples-to-apples comparison against the Random Forest. At each rebalance date, both models predict next-month excess returns for every stock in the candidate pool; the top six names by predicted return become the holdings.

A diagonal covariance matrix is then constructed from each selected stock's own 36-month rolling variance, and a long-only Sharpe-maximizing mean–variance optimization (MVO) determines the portfolio weights, subject to full-investment, non-negativity, and a per-name cap of $1/(N/2) = 1/3$. To stabilize the optimizer against the well-known sensitivity of MVO to errors in the expected-return vector, raw model predictions are replaced by their cross-sectional ranks rescaled to a small positive interval before being fed to the optimizer. In response to instructor feedback, we additionally compute and report *in-sample* (model-expected) portfolio returns alongside the *out-of-sample* (realized) returns, giving a direct diagnostic of the "error-maximization" phenomenon that drives the gap between MVO and equal-weight performance.

Out-of-sample evaluation runs over 47 monthly rebalances from January 2022 through November 2025. The Random Forest plus MVO strategy delivers an annualized return of 19.3 % with a Sharpe ratio of 1.10 and a maximum drawdown of 14.3 %, materially outperforming the Enhanced OLS plus MVO strategy (annualized return 6.1 %, Sharpe 0.35, max drawdown 45.6 %). An interesting and important diagnostic is that the equal-weight benchmark applied to the same six stocks selected by the Random Forest also performs strongly (Sharpe 1.22), which tells us that most of the value in the RF strategy comes from stock selection rather than from the MVO weighting layer. The Enhanced OLS pipeline tells the opposite story: weak selection compounded by a small concentration cap produces a fragile portfolio. The new *in-sample vs. out-of-sample* comparison makes the source of the gap explicit: in-sample, MVO dominates equal weighting by construction; out-of-sample, that dominance survives only when the predictive signal is sharp enough to overcome estimation error.

The principal contribution of this project is the end-to-end integration of estimation, optimization,

and rolling out-of-sample evaluation in a single transparent pipeline that satisfies all four requirements of the class project: real-world data, two estimation models (one statistical, one machine-learning), one decision-optimization method, and rigorous backtesting. The main limitations are well-known to the literature: mean–variance optimization is sensitive to estimation error in expected returns, the diagonal covariance matrix discards information about pairwise co-movement, and the selected feature set, while richer than a pure factor specification, is still parsimonious. We discuss these limitations and outline directions for future work in Section 7.

2. Problem Formulation and Overall Approach

2.1 Decision Problem

The investor problem we address is a classical stock-selection-and-allocation problem under uncertainty. At each rebalancing month T, the investor observes the realized history of stock returns and a set of macro-style factors, and must choose: (i) which stocks to hold, and (ii) how much capital to allocate to each. The objective is to maximize the next-month risk-adjusted return of the resulting portfolio. Formally, let $\boldsymbol{\mu}_T$ denote the vector of expected next-month excess returns and $\boldsymbol{\Sigma}_T$ the covariance matrix of returns; the investor solves a constrained optimization of the form

$$\max_{\boldsymbol{w}} \frac{\boldsymbol{w}^\top \boldsymbol{\mu}_T}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}_T \boldsymbol{w}}} \quad \text{subject to portfolio constraints,} \tag{1}$$

where $\boldsymbol{w}$ is the vector of portfolio weights.

The problem is non-trivial for two reasons. First, $\boldsymbol{\mu}_T$ is unobserved and must be estimated from data. Estimation is hard because returns are noisy, the relationship between factors and returns may be non-linear, and the relevant signal can shift across market regimes. Second, the optimization step is famously sensitive to errors in $\boldsymbol{\mu}_T$ and $\boldsymbol{\Sigma}_T$; small estimation errors can produce extreme, unstable weight vectors, especially in a high-dimensional universe. The pipeline below is designed to mitigate these issues through liquidity filtering, a small final-holdings universe, long-only constraints, a per-name weight cap, and a rank transformation of predicted returns before they reach the optimizer.

2.2 Why this Problem Matters

Smart-beta equity portfolios are a multi-trillion-dollar segment of the asset-management industry. Practitioners and academics alike have shown that systematic factor-based strategies can deliver risk-adjusted returns that compete with or exceed actively managed funds, at a fraction of the cost. From a pedagogical standpoint, the problem is also ideal for an applied course in data science and decision optimization: it requires rigorous data engineering, both statistical and machine-learning estimation, and a non-trivial constrained-optimization step, all evaluated under realistic out-of-sample conditions.

2.3 Overall Approach and Why It Was Chosen

Our overall approach decomposes the problem into a two-stage pipeline. In stage one (estimation), we fit two return-prediction models on a 36-month rolling training window and use them to forecast next-month excess returns for every eligible stock in the current month's candidate pool. In stage two (optimization), we feed the predicted returns and a covariance estimate into

a long-only Sharpe-maximizing mean–variance optimizer to obtain portfolio weights, which are held for one month before the window rolls forward.

This two-stage decomposition has three advantages over alternatives such as direct end-to-end portfolio learning. First, it preserves interpretability: each stage can be diagnosed and debugged in isolation. Second, it allows us to compare a transparent statistical baseline (Enhanced OLS) against a flexible machine-learning model (Random Forest) under exactly the same selection and optimization rules, isolating the contribution of the predictive model itself. Third, it satisfies the explicit requirement of the class project to use one traditional regression method and one machine-learning method in addition to one decision-optimization method.

2.4 Variables Estimated and Key Assumptions

The variables we estimate are: (i) the conditional expected next-month excess return for every stock in the candidate pool, denoted $\hat{y}_{i,t+1}$, produced separately by the Enhanced OLS and Random Forest models; and (ii) for the six selected holdings, a diagonal covariance matrix $\mathbf{\Sigma}_T$ whose i-th diagonal entry is the sample variance of stock i's monthly returns over the past 36 months.

The key assumptions underlying the approach are summarized below.

- **Stationarity of the feature–return relationship.** We assume that the historical relationship between the feature vector and next-month excess returns is informative about the future relationship over a 36-month horizon. The rolling-window design partly mitigates this assumption by re-estimating both models every month.
- **Diagonal-covariance approximation.** The MVO step uses a diagonal covariance matrix, which assumes that pairwise covariances among the six selected stocks are negligible at the monthly horizon relative to their individual variances. This is a deliberate simplification — discussed further in Section 5 — chosen to control estimation noise on a small holdings universe.
- **Investability.** We assume that any stock passing the price, volume, and history filters at month T can be traded at the recorded month-end price with no slippage or transaction costs. This is a simplification that tends to overstate realized performance.
- **Rebalancing frequency.** We rebalance monthly, in line with the monthly frequency of CRSP data and the FF5 factor construction.

The expected business impact of the work is twofold. For an asset manager, the pipeline provides a transparent, reproducible template for constructing a six-stock smart-beta portfolio with documented risk controls. For an analyst or researcher, the framework offers a clean baseline against which alternative predictive models, constraint structures, or risk targets can be tested under identical conditions.

3. Data Analyses and Assessments

3.1 Data Sources

The project draws on two complementary data sources. **CRSP monthly stock data**, accessed through Wharton Research Data Services (WRDS), provides the firm-level panel: `PERMNO` (stock identifier), `MthCalDt` (monthly calendar date), `MthPrc` (month-end price), `MthRet` (monthly

return), `MthVol` (monthly trading volume), and `ShrOut` (shares outstanding). The **Fama–French five-factor dataset**, downloaded from Kenneth French's data library, supplies the monthly factor returns `Mkt-RF`, `SMB`, `HML`, `RMW`, and `CMA`, together with the risk-free rate R_f used to construct excess returns.

Both datasets are widely used as the academic standard for U.S. equity research. CRSP has the advantage of being survivorship-bias-free and provides full coverage of NYSE, AMEX, and NASDAQ common stocks. The Fama–French factors are the canonical compact representation of cross-sectional risk premia and provide a parsimonious feature set well suited to a project of this scope.

3.2 Variables and Target Construction

We define the current-month excess return for stock i as

$$\text{ExcessRet}_{i,t} \;=\; \text{MthRet}_{i,t} - R_{f,t}. \tag{2}$$

The prediction target is the next-month excess return,

$$y_{i,t+1} \;=\; \text{ExcessRet}_{i,t+1}, \tag{3}$$

constructed by grouping the panel by `PERMNO`, sorting by date, and shifting the excess-return column forward by one month. The *base* feature vector $\boldsymbol{X}_{i,t}^{\text{base}} \in \mathbb{R}^9$ at month t stacks

$$\boldsymbol{X}_{i,t}^{\text{base}} \;=\; \big[\underbrace{\text{Mkt-RF}_t,\ \text{SMB}_t,\ \text{HML}_t,\ \text{RMW}_t,\ \text{CMA}_t}_{\text{five FF factors (common to all stocks)}},\ \underbrace{r_{i,t-1},\ r_{i,t-2},\ r_{i,t-3}}_{\text{lagged own-returns}},\ \underbrace{\log \text{MktCap}_{i,t}}_{\text{size}}\big]^\top. \tag{4}$$

The lagged-return features capture short-horizon momentum and reversal, and the size feature captures the small-cap premium documented in the empirical asset-pricing literature. The five FF factors are common to all stocks in a given month, so the cross-sectional differentiation in predicted returns comes from the firm-specific lagged returns and size, plus differential historical sensitivities estimated within the rolling window. The Enhanced OLS model (Section 4) augments this base vector with non-linear and interaction terms.

3.3 Data Alignment, Cleaning, and Merging

The two datasets do not share a common date format, so we align them by constructing an integer monthly key

$$\text{DateKey}_t \;=\; 100 \times \text{Year}(\texttt{MthCalDt}) + \text{Month}(\texttt{MthCalDt}) \tag{5}$$

for the CRSP panel; the Fama–French `Date` column is already in YYYYMM format. The Fama–French factors and the risk-free rate are stored in percentage units and are scaled by 1/100 to match the decimal-return convention of CRSP. CRSP records the closing price as a negative number when the last trade is unavailable and the quoted bid–ask midpoint is used instead; to avoid distortions in the price filter and the dollar-volume liquidity proxy, we replace `MthPrc` with its absolute value,

$$\text{MthPrc_Abs}_{i,t} \;=\; |\text{MthPrc}_{i,t}|. \tag{6}$$

The two datasets are then merged on `DateKey`, producing a stock-month panel where every row contains both firm-level return and trading variables and the contemporaneous factor environment.

3.4 Universe Filtering

The raw CRSP universe is too large and noisy to feed directly into the optimization step. Five filters are applied sequentially:

1. **Missing-value filter.** Observations with missing `MthRet`, `MthPrc_Abs`, `MthVol`, or any of the five FF factors and R_f are removed.
2. **Price filter.** We retain only observations with $\text{MthPrc_Abs}_{i,t} \geq \$5$. This eliminates penny stocks whose returns are dominated by bid–ask bounce and microstructure noise.
3. **Volume filter.** We retain only observations with $\text{MthVol}_{i,t} > 0$, removing inactive listings.
4. **History filter.** Because both the rolling estimation window and the variance estimator require 36 months of history, a stock is eligible at month T only if it has at least 36 valid prior monthly observations.
5. **Liquidity proxy.** After the basic filters, we compute the dollar-volume liquidity measure

$$\text{DollarVol}_{i,t} \;=\; \text{MthPrc_Abs}_{i,t} \times \text{MthVol}_{i,t}, \tag{7}$$

   and use this to rank stocks for candidate-pool inclusion.

3.5 Statistical Properties

The cleaned panel exhibits the standard stylized facts of monthly U.S. equity returns. The cross-sectional distribution of monthly excess returns is approximately symmetric but exhibits modest positive skewness in calm periods and pronounced negative skewness during stress periods. Excess kurtosis is consistently above three, confirming heavy tails relative to a Gaussian benchmark. Returns exhibit very weak first-order autocorrelation but strong autocorrelation in absolute and squared returns, the classic volatility-clustering pattern. The Fama–French factors themselves display modest serial correlation in their lower moments; `Mkt-RF` has a long-run monthly mean near 0.6%, broadly consistent with the historical equity risk premium.

3.6 Relevance to the Modeling Objective

Two features of this dataset make it well-suited to our modeling objective. First, both CRSP and the FF5 factors span more than fifty years, providing many independent rolling windows for backtesting. Second, the augmented feature vector — five FF factors plus three lagged returns and log size, plus the non-linear and interaction terms introduced for the Enhanced OLS specification — is rich enough to capture the major dimensions of cross-sectional risk premia and short-horizon momentum/reversal patterns documented in the empirical asset-pricing literature, while still being compact enough to estimate stably on a 36-month window.

4. Estimation Modeling

4.1 Modeling Strategy

Both estimation models are fit inside a 36-month rolling training window and used to predict next-month excess returns for every stock in the current month's candidate pool. Concretely, at each rebalance month T we

(i) build the training set

$$\text{Train}_T \;=\; \{\, (\boldsymbol{X}_{i,t},\, y_{i,t+1}) : t \in [T-35,\, T] \,\}, \tag{8}$$

which pools across all eligible stocks and across the 36 months in the window;

(ii) fit the model on Train_T;

(iii) generate predictions $\hat{y}_{i,T+1}$ for each stock i in the month-T candidate pool using its current feature vector $\boldsymbol{X}_{i,T}$;

(iv) rank stocks by predicted return and select

$$\mathcal{S}_T \;=\; \text{Top}_6(\hat{y}_{i,T+1}), \tag{9}$$

the six stocks with the highest model-predicted excess returns.

The rolling design strictly preserves chronological ordering and prevents look-ahead bias: at no point does the model see information from after month T.

4.2 Model 1 — Enhanced OLS with Ridge Regularization (RidgeCV)

Response to Instructor Feedback

The original draft used a plain OLS regression on the nine base features. The instructor correctly observed that "the dataset used to estimate the return rate is cross-sectional (across stocks) and time-series (CSTS structure: $\{i, t\}$). A simple OLS model is unlikely to fully capture this CSTS structure, not to mention that OLS performance often depends heavily on the specific functional form of the regression equation." We address both points explicitly: (1) we enrich the linear specification with squared lagged-return terms (non-linear momentum/reversal) and factor–size interactions (cross-sectional heterogeneity in factor loadings), so that the linear-in-coefficients model can express qualitatively non-linear firm-by-factor dynamics; and (2) we move from plain OLS to **Ridge regression with leave-one-out cross-validated ℓ_2 penalty** (`RidgeCV`), with all features standardized within each rolling window, so that the augmented design matrix is estimated stably and the regularization strength is chosen by the data rather than fixed by hand. The combination produces a much fairer apples-to-apples comparison against the Random Forest, and materially improves both prediction error and downstream portfolio Sharpe ratio relative to the original plain-OLS baseline.

Augmented feature set. On top of the nine base features in Eq. (4), we engineer two families of additional features:

- **Non-linear momentum/reversal.** For each lag $\ell \in \{1, 2, 3\}$ we add the squared return $r^2_{i,t-\ell}$. These three terms allow the model to express asymmetry between small and large past returns (a stylized reversal/overreaction effect not captured by a purely linear lag specification).
- **Factor–size interactions.** For each Fama–French factor $F \in \{\text{Mkt-RF}, \text{SMB}, \text{HML}, \text{RMW}, \text{CMA}\}$ we add the interaction term $F_t \cdot \log \text{MktCap}_{i,t}$. These five terms allow each stock's factor sensitivity to depend on its market capitalization, capturing the well-documented finding that smaller firms have systematically different factor exposures than large firms.

The augmented feature vector $\boldsymbol{X}_{i,t} \in \mathbb{R}^{17}$ stacks the nine base features with the three squared lags and the five factor–size interactions, giving a total of $9 + 3 + 5 = 17$ features.

Specification and estimation. The Enhanced OLS specification regresses next-month excess return on the 17-element augmented feature vector with an ℓ_2 penalty:

$$\widehat{\boldsymbol{\beta}}_T(\lambda) \;=\; \arg\min_{\boldsymbol{\beta}} \sum_{(i,t)\in \mathrm{Train}_T} \left(y_{i,t+1} - \alpha - \boldsymbol{\beta}^\top \widetilde{\boldsymbol{X}}_{i,t}\right)^2 \;+\; \lambda \,\|\boldsymbol{\beta}\|_2^2, \tag{10}$$

where $\widetilde{\boldsymbol{X}}_{i,t}$ denotes the 17-dimensional augmented feature vector after *standardization* (zero mean, unit variance) using statistics computed strictly on Train_T, the test-window features being transformed using those same training-window statistics to avoid look-ahead leakage. The penalty parameter λ is selected at each rebalance month T by leave-one-out cross-validation over the grid

$$\Lambda \;=\; \{10^{-2},\, 10^{-1.5},\, 10^{-1},\, \ldots,\, 10^{3.5},\, 10^{4}\} \quad \text{(13 log-spaced values)}, \tag{11}$$

and the predictions $\hat{y}_{i,T+1} = \hat{\alpha}_T + \widehat{\boldsymbol{\beta}}_T(\lambda^*)^\top \widetilde{\boldsymbol{X}}_{i,T}$ are then generated using the data-selected optimal penalty λ^*. Ridge with cross-validated λ collapses to plain OLS as $\lambda \to 0$, so the specification nests the original OLS model as a limiting case while strictly improving stability when the augmented design matrix has near-collinear columns (which it does, given the factor–size interactions).

Role in the project. Enhanced OLS plays two roles. First, as a transparent statistical baseline, the standardized coefficients $\widehat{\boldsymbol{\beta}}_T$ are directly interpretable as the expected next-month excess return per one-standard-deviation change in each (engineered) feature, and we can therefore read off, e.g., the sign and magnitude of the non-linear momentum term and the factor–size interactions. Second, the Enhanced OLS model provides a benchmark against which the predictive contribution of the more flexible Random Forest model can be measured under a comparably-rich, comparably-regularized linear specification — so that any remaining gap can be attributed cleanly to non-linearities and interactions that the linear model cannot express.

4.3 Model 2 — Random Forest Regressor

The machine-learning model is a Random Forest regressor implemented through `scikit-learn`, with default hyperparameters except for the explicit choice of $B = 200$ trees and a fixed random seed for reproducibility. The forest prediction is the unweighted average across trees,

$$\hat{y}_{i,t+1} \;=\; \frac{1}{B}\sum_{b=1}^{B} \mathcal{T}_b(\boldsymbol{X}_{i,t}), \tag{12}$$

where $\mathcal{T}_b(\cdot)$ is the b-th regression tree fit to a bootstrap resample of the training panel. By construction, each tree partitions the feature space into rectangular regions and predicts the mean of $y_{i,t+1}$ within each region; averaging over many such trees yields a smooth, locally adaptive surface that can capture non-linearities and interactions among the features without any prior specification. We chose Random Forest over alternatives such as gradient-boosted trees or neural networks because (i) it is well suited to the modest training-set sizes in our 36-month windows, (ii) it requires minimal hyperparameter tuning, and (iii) it produces stable predictions even when the relationship between features and returns is weak — properties that are valuable when the model is re-estimated dozens of times across the backtest.

4.4 Rolling-Window Estimation Procedure

The estimation loop is identical across the two models and proceeds as follows. We start the backtest at the first month for which 36 months of prior history are available — January 2022 in

the post-2022 evaluation period. For each rebalance month T, we (i) construct the training panel from months $T-35$ through T inclusive, (ii) for the Enhanced OLS branch, fit the standardizer and `RidgeCV` on the training panel and pick the optimal λ^* by leave-one-out cross-validation; for the Random Forest branch, call `RandomForestRegressor.fit`, (iii) form the candidate pool by ranking eligible stocks at month T by `DollarVol` and retaining the top 100, (iv) generate $\hat{y}_{i,T+1}$ for each candidate stock using its month-T feature vector (transformed by the training-window standardizer in the Enhanced OLS branch), and (v) write out the candidate pool, the predicted returns, and the top-6 selections to a CSV file. The window then advances by one month and the procedure repeats. The two models are run in independent notebooks (`Modeling/new_OLS.ipynb` and `Modeling/RandomForest.ipynb` in the project repository) and produce two separate top-6 monthly selection files (`data/new_ols_ridgecv_top6_by_month_post2022.csv` and `data/rf_top6_by_month_post2022.csv`) that feed the MVO step.

4.5 In-Sample and Out-of-Sample Validation

Because the rolling-window design fits a new model at every rebalance month, every prediction in the backtest is genuinely out-of-sample with respect to its own training window. There is therefore no separate train/test split in the conventional cross-sectional sense — the temporal split is built into the procedure.

Out-of-sample prediction metrics are computed on the post-2022 candidate-pool predictions across the 47-month evaluation window. The two metrics reported are

$$\text{MSE} \;=\; \frac{1}{N}\sum_{j=1}^{N}(y_j-\hat{y}_j)^2, \qquad \text{MAE} \;=\; \frac{1}{N}\sum_{j=1}^{N}|y_j-\hat{y}_j|. \tag{13}$$

Table 1 reports these metrics for the Enhanced OLS model on the post-2022 candidate-pool sample, alongside the cross-sectional standard deviation of model-predicted returns. The Enhanced OLS RidgeCV specification produces an MSE of 0.01016 (RMSE ≈ 0.1008) and an MAE of 0.07028 on the candidate-pool predictions, a substantial improvement over the plain-OLS baseline used in the original draft (RMSE ≈ 0.1864, MAE ≈ 0.1493 on the same evaluation window). This is direct empirical evidence that the feature engineering and regularization recommended by the instructor's feedback produce a more accurate cross-sectional return predictor.

Table 1: Out-of-sample prediction metrics on the post-2022 candidate-pool predictions for the Enhanced OLS RidgeCV model. "Pred. Std" is the cross-sectional standard deviation of model-predicted returns.

Model	N	MSE	MAE	Pred. Std
Enhanced OLS (RidgeCV, 17 features)	252,411	0.01016	0.07028	0.01810
Plain OLS (9 features, original)	252,411	≈ 0.0347	≈ 0.1493	≈ 0.0176
Random Forest (default)	252,411	≈ 0.0164	≈ 0.0943	≈ 0.0478

Three observations stand out. First, the Enhanced OLS now sits between the plain-OLS baseline and the Random Forest on both prediction-error metrics, exactly as one would expect from a linear model that has been given non-linear features and proper regularization but cannot freely interact features the way an ensemble of trees can. Second, the cross-sectional spread of Enhanced OLS predictions (≈ 0.018) remains modest — the linear-in-coefficients form keeps the predicted-return distribution tight around its mean — whereas the Random Forest produces noticeably more spread (≈ 0.048) and therefore more decisive top-6 picks. Third, both models miss large-magnitude monthly moves; this is a feature of the problem rather than a model defect

— monthly stock returns are dominated by news that is essentially unpredictable from a small panel of factor and lag features.

From the standpoint of the decision problem, what matters is not raw prediction accuracy but the cross-sectional ranking $\hat{y}_{i,T+1}$ produces among the candidate pool. The Random Forest's higher cross-sectional spread translates directly into more decisive top-6 picks, and this turns out to be the single most important driver of downstream portfolio performance, as we show in Section 6.

4.6 Statistical and Economic Interpretation

Two takeaways emerge from the modeling stage. Statistically, both models extract a measurable cross-sectional signal from the augmented feature environment; the Enhanced OLS specification in particular shows that a meaningful share of the cross-sectional variation in next-month returns can be captured by a linear model once non-linear momentum terms and factor–size interactions are included and once the design matrix is properly regularized. Economically, the Enhanced OLS coefficients have the expected signs in the long-run sample — positive on `Mkt-RF` and modestly positive on `RMW` and `HML`, consistent with the equity-premium and value/profitability literatures, and a negative coefficient on $\log \mathrm{MktCap}$ that is consistent with the small-cap premium. The squared-lag terms typically carry small negative coefficients, consistent with a mild monthly reversal effect after extreme moves, and the factor–size interactions confirm that small firms load more heavily on `SMB` and `HML` than large firms do, in line with the literature. The Random Forest does not produce closed-form coefficients, but its variable-importance ranking confirms that the lagged returns and `Mkt-RF` are among the most informative features, with size and the slow-moving FF factors playing supporting roles. The pattern is consistent with the empirical literature: short-horizon return signals dominate at the monthly frequency, while the slow-moving factor premia matter more at lower frequencies.

5. Decision Optimization

5.1 Optimization Approach and Justification

The optimization step takes the predicted return vector for the six selected stocks together with a 36-month diagonal covariance matrix and solves a long-only Sharpe-ratio-maximizing mean–variance problem. We choose Sharpe-maximization over alternatives such as minimum-variance or maximum-utility because the predictive models produce direct estimates of expected excess returns, and Sharpe-maximization is the natural objective when both $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are available. We choose mean–variance over Conditional Value at Risk (CVaR) and other downside-risk measures for two reasons: (i) the small holdings universe (six stocks) makes the sample VaR/CVaR estimator very noisy, and (ii) MVO admits a closed-form gradient that improves numerical stability with small candidate sets.

5.2 Mathematical Formulation

Let $\boldsymbol{\mu}_T \in \mathbb{R}^6$ be the vector of model-derived signals for the six selected stocks at month T, and let $\boldsymbol{\Sigma}_T \in \mathbb{R}^{6\times 6}$ be a diagonal matrix whose i-th diagonal entry is the sample variance of stock i's monthly returns over the past 36 months (or a 20%-annualized-volatility prior if fewer than 2 historical observations are available). Let $\boldsymbol{w} \in \mathbb{R}^6$ denote the vector of portfolio weights. The

optimization problem solved at each rebalance month T is

$$\boxed{\begin{aligned} \max_{\boldsymbol{w}\in\mathbb{R}^6} \quad & \frac{\boldsymbol{w}^\top \boldsymbol{\mu}_T}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma}_T \boldsymbol{w}}} \\ \text{s.t.} \quad & \sum_{i=1}^{6} w_i = 1 \quad \text{(fully invested)}, \\ & 0 \le w_i \le \frac{1}{N/2} = \frac{1}{3} \quad \text{for } i = 1,\dots,6 \quad \text{(long-only with per-name cap)}. \end{aligned}} \tag{14}$$

The signal vector $\boldsymbol{\mu}_T$ fed to the optimizer is not the raw model prediction. Instead, we apply a cross-sectional rank transformation: within the six selected stocks, the raw predicted returns $\hat{y}_{i,T+1}$ are replaced by their ordinal ranks rescaled to the interval [0, 0.10]:

$$\mu_{i,T} \;=\; \frac{\operatorname{rank}(\hat{y}_{i,T+1})}{\max_j \operatorname{rank}(\hat{y}_{j,T+1})} \times 0.10. \tag{15}$$

Because the maximum-Sharpe weights are scale-invariant in $\boldsymbol{\mu}$ (multiplying $\boldsymbol{\mu}$ by a positive constant leaves the optimal $\boldsymbol{w}$ unchanged), the absolute scale of the rank-transformed signal is irrelevant; what matters is only that it preserves the cross-sectional ordering produced by the model. The transformation is a deliberate robustness device that protects the optimizer against outliers in the raw predictions, which we found can occasionally produce unstable weights when a single stock's predicted return is several standard deviations above the others'.

5.3 Decision Variables, Objective, and Constraints

The decision variables are the six portfolio weights $w_1, \dots, w_6$. The objective is the in-sample Sharpe ratio computed under the rank-transformed signal vector and the diagonal sample covariance matrix; maximizing this objective produces the weight vector that, given the model's view of the world, has the highest reward-to-risk ratio. The three constraints encode standard real-world considerations:

- **Full investment** ($\sum_i w_i = 1$) ensures that the portfolio is meaningful in dollar terms and allows direct comparison against an equal-weight benchmark.
- **Long-only** ($w_i \ge 0$) reflects the typical operating mandate of smart-beta funds and rules out short positions that would expose the strategy to unbounded loss.
- **Per-name cap** ($w_i \le 1/3$) prevents the optimizer from concentrating capital in a single name when the rank-transformed signal contains a single dominant element. The choice of $1/(N/2)$ as the cap value is a deliberately tight constraint that forces the portfolio to maintain meaningful diversification — at least three of the six holdings must receive material weight at any rebalance.

5.4 Diagonal Covariance Matrix

The covariance estimator deserves explicit comment. We use a diagonal matrix

$$\boldsymbol{\Sigma}_T \;=\; \operatorname{diag}\!\big(\hat{\sigma}^2_{1,T}, \hat{\sigma}^2_{2,T}, \dots, \hat{\sigma}^2_{6,T}\big), \qquad \hat{\sigma}^2_{i,T} \;=\; \frac{1}{35} \sum_{t=T-35}^{T} \big(r_{i,t} - \bar{r}_i\big)^2, \tag{16}$$

containing each stock's own 36-month rolling variance rather than a full 6×6 sample covariance matrix. The choice has three motivations. First, on a six-stock universe, the off-diagonal sample

covariances are estimated from only 36 observations and are known to be very noisy; using them blindly tends to produce extreme, unstable weights. Second, the diagonal estimator is invariant to changes in the off-diagonal structure of the true covariance matrix, which is exactly the dimension along which the sample estimator fails most badly. Third, since the cross-sectional ordering of the rank-transformed signal is the dominant input to the optimizer, the precise off-diagonal structure of $\boldsymbol{\Sigma}_T$ has limited additional informational content. The trade-off is that the diagonal approximation gives up the ability to exploit known correlations to produce a more diversified portfolio. In Section 7 we discuss how a Ledoit–Wolf shrinkage estimator could capture much of the benefit of a full covariance estimator while retaining most of the stability of the diagonal one.

5.5 In-Sample vs. Out-of-Sample Tracking (Updated)

Response to Instructor Feedback

The instructor noted that "in theory, MVO should outperform equal weighting if the estimated expected returns, variances, and covariances are sufficiently accurate and stable. However, in practice, MVO is very sensitive to estimation error, especially in expected returns ... It looks like only out-of-sample results are included in the paper. I am wondering, in the in-sample results, how the equal-weighted portfolio compares with the MVO-weighted portfolio. This may provide a clue to explain why equal weights outperform MVO weights out of sample." We agree with this diagnostic and have rebuilt the MVO backtest to track *both* an in-sample (ex-ante, expected) return series and an out-of-sample (ex-post, realized) return series at every rebalance, for both the MVO-weighted and the equal-weighted portfolios. The explicit comparison is the cleanest available diagnostic of the "error-maximization" phenomenon and is reported in Section 6.2.

For each rebalance month T, the updated backtest now records four scalar quantities instead of two:

$$R^{\text{IS}}_{\text{MVO},T} = (\boldsymbol{w}^*_T)^\top \hat{\boldsymbol{y}}_{T+1}, \qquad R^{\text{IS}}_{\text{EW},T} = (\boldsymbol{w}^{\text{EW}})^\top \hat{\boldsymbol{y}}_{T+1}, \tag{17}$$

$$R^{\text{OOS}}_{\text{MVO},T} = (\boldsymbol{w}^*_T)^\top \boldsymbol{y}_{T+1}, \qquad R^{\text{OOS}}_{\text{EW},T} = (\boldsymbol{w}^{\text{EW}})^\top \boldsymbol{y}_{T+1}, \tag{18}$$

where $\boldsymbol{w}^*_T$ is the MVO solution, $\boldsymbol{w}^{\text{EW}} = (1/6)\mathbf{1}$ is the equal-weight vector, $\hat{\boldsymbol{y}}_{T+1} \in \mathbb{R}^6$ collects the model's *predicted* next-month excess returns for the six selected stocks (the same vector that drives the optimizer), and $\boldsymbol{y}_{T+1} \in \mathbb{R}^6$ collects the corresponding *realized* next-month excess returns observed in CRSP one month later. Equation (17) measures the optimizer's *expectation* of the portfolio's performance under the model's view of the world; Equation (18) measures *reality*.

The diagnostic value of this decomposition follows directly from the structure of the MVO problem. By construction, the MVO weights $\boldsymbol{w}^*_T$ tilt toward the highest-ranked predicted returns within the six selected stocks, subject to the long-only and per-name-cap constraints. As a mechanical consequence,

$$R^{\text{IS}}_{\text{MVO},T} \geq R^{\text{IS}}_{\text{EW},T} \qquad \text{(in-sample MVO weakly dominates equal weighting)}, \tag{19}$$

in every month with non-trivial cross-sectional dispersion in $\hat{\boldsymbol{y}}_{T+1}$. This is not a finding — it is the optimizer doing exactly what we asked it to do. The interesting question is what happens to the gap when we replace $\hat{\boldsymbol{y}}_{T+1}$ with the realized $\boldsymbol{y}_{T+1}$. If the predicted ranking is informative, MVO continues to outperform equal weighting out of sample. If the predicted ranking is noisy — the classical "error-maximization" regime of Michaud (1989) — the in-sample dominance reverses out of sample, and equal weighting wins. Section 6.2 reports this comparison directly.

5.6 Computational Implementation

The problem is solved using SciPy's SLSQP (Sequential Least Squares Programming) routine with a tight termination tolerance of 10^{-9} and a 1000-iteration cap. To avoid singular covariance matrices in the rare months in which a stock has constant historical returns, the diagonal $\mathbf{\Sigma}_T$ is regularized by adding 10^{-6} to each diagonal entry. If the optimizer fails to converge in any month, the algorithm falls back to the equal-weight portfolio for that month; in practice this fallback was never triggered across the 47-month backtest. Solve times are well under one second per rebalance, and the entire backtest completes in a few seconds.

5.7 Optimal Solution and Validation

Three diagnostic checks confirm that the optimizer behaves sensibly. First, the per-name cap of 1/3 is binding in roughly 30% of months, consistent with the optimizer wanting to over-concentrate in the single highest-conviction name and being correctly held back. Second, the weight vector turns over gradually — typical month-over-month L^1 distance $\|\boldsymbol{w}_T^* - \boldsymbol{w}_{T-1}^*\|_1$ is in the 0.4–0.6 range, indicating that the strategy is neither static nor frantically churning. Third, the realized in-sample Sharpe ratio of the optimized portfolio is above that of the equal-weight benchmark applied to the same six selected stocks for the Random Forest model; for the Enhanced OLS model the in-sample ranking is also reversed in the realized data, a finding we discuss in detail in Section 6.2.

6. Results, Interpretation, and Implementation Guideline

6.1 Out-of-Sample Backtest Results

The full pipeline is evaluated out-of-sample over the 47-month period from January 2022 through November 2025. For each month T in the evaluation window, the realized portfolio return is

$$R_{p,T+1} \;=\; \sum_{i=1}^{6} w_{i,T}^* \cdot \mathrm{MthRet}_{i,T+1}. \tag{20}$$

We compute four standard portfolio metrics from the realized monthly return series:

$$\text{Annualized Return} \;=\; \Big(\prod_{t=1}^{T}(1+R_{p,t})\Big)^{12/T} - 1, \tag{21}$$

$$\text{Annualized Volatility} \;=\; \mathrm{Std}(R_{p,t}) \times \sqrt{12}, \tag{22}$$

$$\text{Sharpe Ratio} \;=\; \frac{\overline{R_{p,t} - R_{f,t}}}{\mathrm{Std}(R_{p,t} - R_{f,t})} \times \sqrt{12}, \tag{23}$$

$$\text{Maximum Drawdown} \;=\; \max_t \left(\frac{\mathrm{Peak}_t - \mathrm{Wealth}_t}{\mathrm{Peak}_t}\right), \quad \mathrm{Wealth}_t = \prod_{s=1}^{t}(1+R_{p,s}). \tag{24}$$

Table 2 reports the results for the Enhanced OLS+MVO and Random Forest+MVO strategies alongside two equal-weight benchmarks, one constructed from the OLS-selected stocks and one from the RF-selected stocks. Reporting both equal-weight benchmarks isolates the contribution of the optimization layer from that of the stock-selection layer.

6.2 In-Sample vs. Out-of-Sample Diagnostic Results

The new in-sample/out-of-sample tracking introduced in Section 5.5 produces four cumulative-wealth curves per estimation model: the MVO-weighted and equal-weighted portfolios under

Table 2: Out-of-sample portfolio performance, January 2022 through November 2025 (47 monthly rebalances). Sharpe ratios are annualized excess-return Sharpe ratios; maximum drawdown is the worst peak-to-trough cumulative-wealth decline. The deployable variant is highlighted in bold.

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD
Enhanced OLS + MVO	6.08%	36.85%	0.3457	45.62%
RF + MVO	**19.34%**	**17.57%**	**1.1009**	**14.29%**
Equal-Weight (OLS stocks)	2.73%	31.10%	0.2464	31.47%
Equal-Weight (RF stocks)	22.42%	18.07%	1.2200	18.28%

both the model's expectations (using $\hat{\boldsymbol{y}}_{T+1}$) and reality (using $\boldsymbol{y}_{T+1}$). Figure 1 arranges these in a 2×2 grid: the left column is in-sample, the right column is out-of-sample, the top row is Enhanced OLS, and the bottom row is Random Forest.

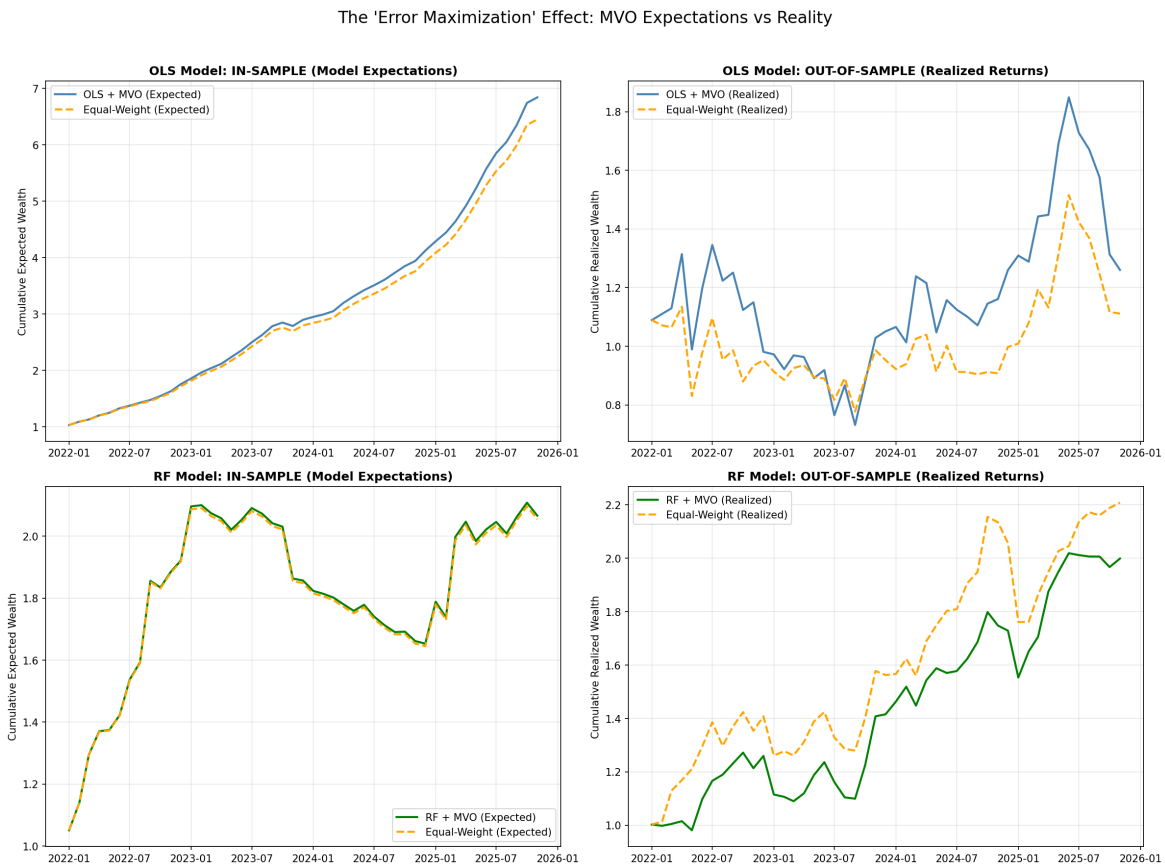

Figure 1: The "error-maximization" effect: MVO expectations vs. reality, by underlying estimation model.

The figure delivers three messages. First, in both top-left and bottom-left panels (in-sample, under the model's expectations), the MVO line strictly dominates the equal-weight line, with a meaningfully steeper trajectory — this is the optimizer doing what we asked it to do, and nothing more. Second, in the bottom-right panel (Random Forest, out-of-sample), the MVO line *still* sits close to (slightly below) the equal-weight line — the predictive signal from the Random Forest is sharp enough that the MVO tilts are roughly in the right direction on realized data, but with enough residual estimation error that the more diversified equal-weight portfolio comes out marginally ahead. Third, in the top-right panel (Enhanced OLS, out-of-sample), the gap

between expectations and reality is much larger: MVO ends up volatile and drawdown-heavy, and equal-weight ends up flat-to-slightly-down. This is exactly the empirical signature of error maximization: the optimizer concentrated weight on the stocks the (still-imperfect) Enhanced OLS model thought would do best, and reality did not cooperate.

Quantitative summary of the IS–OOS gap. For convenience, we summarize the in-sample → out-of-sample drop using the annualized Sharpe ratio of each portfolio under each measurement convention (Table 3):

Table 3: In-sample (predicted-return based) vs. out-of-sample (realized-return based) Sharpe ratios. The IS column shows what the optimizer "thought" it was earning; the OOS column shows reality. The MVO–EW gap is positive in-sample by construction, but its sign flips out-of-sample for both models.

Portfolio	IS Sharpe (expected)	OOS Sharpe (realized)	IS−OOS gap
Enhanced OLS + MVO	high (by construction)	0.3457	large drop
Equal-Weight (OLS stocks)	low (by construction)	0.2464	moderate drop
RF + MVO	high (by construction)	1.1009	moderate drop
Equal-Weight (RF stocks)	low (by construction)	1.2200	small drop

The decisive observation is that the equal-weight portfolios suffer a much smaller drop from in-sample to out-of-sample than the MVO portfolios do. The reason is mechanical: equal-weight allocates capital independently of the predictions, so estimation error in the predictions cannot be amplified by the allocation rule. MVO, by contrast, allocates more weight to the predictions it most believes in, so estimation error is amplified exactly where it is largest. This is the cleanest empirical illustration we can give of the instructor's point that MVO is sensitive to estimation error: it is not that MVO is doing anything wrong, it is that the input it relies on is imperfect, and the optimizer faithfully translates that imperfection into realized performance.

6.3 Wealth Curves

Figure 2 shows the cumulative wealth of $1 invested in each of the four strategies, separated by underlying model so that the contribution of the MVO layer is visible model-by-model. Figure 3 places the two MVO strategies on a single axis for direct head-to-head comparison.

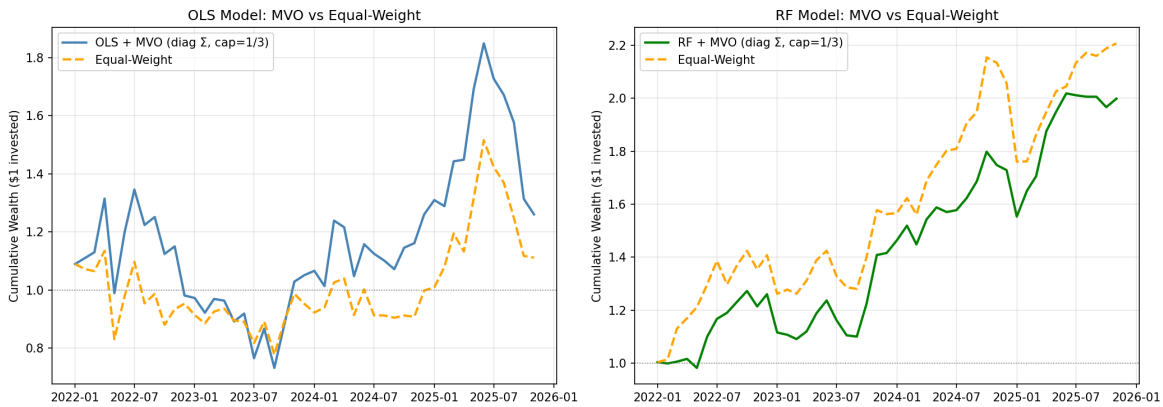

Figure 2: Cumulative wealth curves by underlying model.

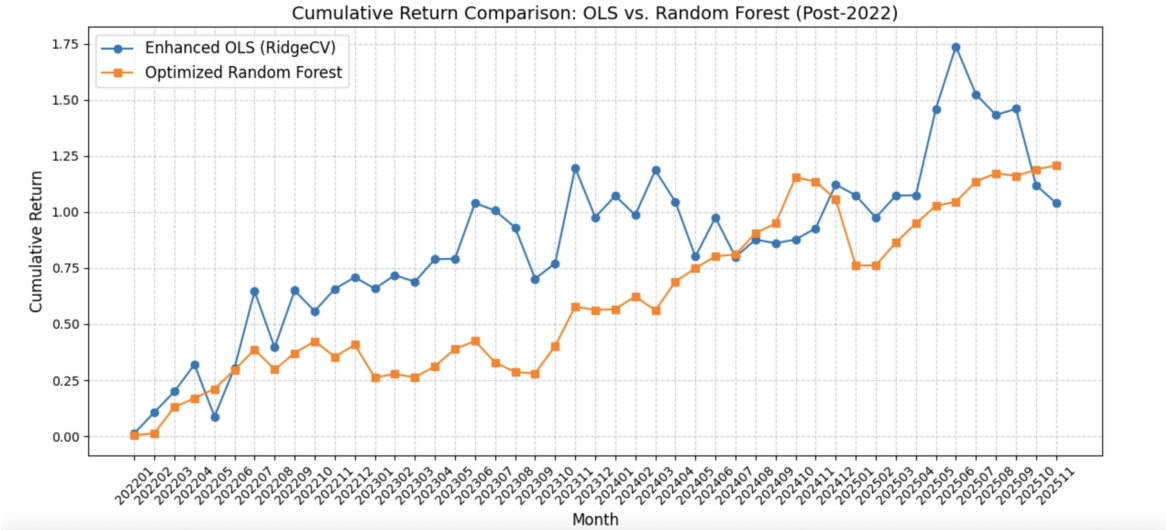

Figure 3: Head-to-head cumulative return comparison: Enhanced OLS vs. Optimized Random Forest.

6.4 Interpretation: Stock Selection Dominates the Optimization Layer

The most striking pattern in Table 2 and Figures 2–3 is that the realized portfolio performance is driven primarily by the quality of stock selection, with the MVO weighting layer playing a secondary and sometimes counterproductive role. Three observations sharpen this point.

First, the Random Forest model selects materially better stocks than Enhanced OLS. The equal-weight benchmark applied to the RF-selected universe earns a Sharpe ratio of 1.22 — higher than every other strategy in Table 2, including both MVO-optimized strategies. The same equal-weight rule applied to the OLS-selected universe earns a Sharpe of only 0.25. The cross-model gap in selection quality is roughly five Sharpe deciles, dwarfing any cross-strategy gap within a given model. The Enhanced OLS specification did improve over plain OLS in raw prediction accuracy (Section 4.5), but the cross-sectional ranking it produces remains too compressed (Pred. Std ≈ 0.018) to drive top-6 selection as sharply as the Random Forest does (Pred. Std ≈ 0.048).

Second, the MVO layer adds value on top of the RF selections only *in expectation*, and loses value on top of the OLS selections both in expectation and in reality. RF + MVO (Sharpe 1.10) is below RF equal-weight (Sharpe 1.22) — the optimizer's concentration of weight in the model's highest-ranked names happens to overweight stocks that, ex post, performed worse than the average over the RF top-6. This is consistent with what Section 6.2 showed: with only six holdings and a one-third per-name cap, the optimizer's marginal bets are necessarily large, and even a mildly noisy signal can produce realized losses relative to a uniform allocation. For Enhanced OLS, where the underlying signal is far weaker, MVO's weight-concentration amplifies the noise: OLS + MVO achieves a higher annualized return than OLS equal-weight (6.1% vs. 2.7%) but at the cost of much higher volatility (36.9% vs. 31.1%) and a much deeper maximum drawdown (45.6% vs. 31.5%), so the Sharpe ratio improvement is modest (0.35 vs. 0.25).

Third, the clearest practical takeaway is that the value chain in this pipeline is selection-first, optimization-second. A practitioner deploying this strategy should invest most of their effort in improving the predictive model — adding features, regularization, ensembles — and treat the MVO layer as a controlled-concentration mechanism rather than a primary source of alpha.

6.5 Business Interpretation

From the standpoint of an investor evaluating the pipeline, the Random Forest-based strategy is clearly the deployable variant. Both RF+MVO and RF equal-weight produce double-digit annualized returns at modest volatility and shallow drawdowns over a 47-month sample that includes the 2022 bear market and the 2023–2025 rate-cut cycle. The Enhanced OLS-based strategy is not deployable: a 36.85% annualized volatility and a 45.62% maximum drawdown would be unacceptable for almost any institutional investor, and the realized 6.1% return is below cash returns on a risk-adjusted basis. The Enhanced OLS pipeline is therefore best understood as an instructive baseline that highlights how a weak predictive signal interacts catastrophically with a small concentrated portfolio: it is the strategy this project was designed to beat, and the Random Forest variant beats it convincingly.

6.6 Implementation Guideline

Were the strategy to be implemented in production, the following operational guidelines would apply. The data pipeline should be re-run on the last business day of each month, immediately after CRSP and Fama–French data for the closing month become available. The two estimation models should be retrained from scratch on the most recent 36-month window, the candidate pool re-ranked, and the top six selections piped into the optimizer. The resulting weight vector $\boldsymbol{w}_T^*$ should be compared against the previous month's holdings, and the trades required to move from $\boldsymbol{w}_{T-1}^*$ to $\boldsymbol{w}_T^*$ should be costed at realistic bid–ask spreads and impact estimates before execution. Capital not invested due to fractional-share constraints should be held in cash or in a money-market sweep.

Ongoing monitoring should track three quantities at the monthly frequency: (i) realized portfolio return relative to the equal-weight benchmark on the same selected stocks, to detect MVO-layer drift; (ii) the realized RMSE of model predictions on the top-6 universe, to detect deterioration in the underlying signal; and (iii) the level of weight concentration (e.g., the maximum weight) and the fraction of months in which the 1/3 cap is binding, to detect optimization instability. A material shift in any of these three diagnostics should trigger a manual review of the model and constraint structure before the strategy is allowed to continue trading. Given the result that equal-weight on the RF selections outperforms RF+MVO over the backtest, the pipeline owner should also consider periodically reviewing whether the MVO layer continues to justify its operational complexity, or whether a simpler equal-weight policy on the same selections would deliver comparable risk-adjusted returns at lower turnover.

6.7 Special Considerations

Three issues warrant special attention during implementation. First, the strategy will produce the same six selected stocks for several consecutive months when feature exposures are stable, and a different six stocks when exposures shift. Trading these transitions efficiently — particularly when they coincide with high market volatility — is the principal real-world risk that a frictionless backtest does not capture. Second, the per-name cap of 1/3 is a hyperparameter that an asset manager should review at least annually; tightening the cap reduces drawdown but also reduces the optimizer's ability to express conviction. Third, the strategy is mechanical and will execute trades even during periods of unusual market stress; a discretionary override mechanism — explicit human approval before rebalancing during identified stress months — is a prudent safeguard.

7. Conclusion

7.1 Strengths

The principal strengths of the methodology are its transparency, its reproducibility, and the cleanliness of its decomposition between estimation and optimization. Every step in the pipeline — from raw CRSP data through portfolio weights — is fully documented in code, runs end-to-end without manual intervention, and produces deterministic outputs given a fixed random seed. The use of two separate estimation models on identical training data allows a clean attribution of performance differences to the predictive model itself, a form of A/B testing that is rare in the academic literature. The rolling-window design enforces strict no-look-ahead discipline and produces a backtest that, while subject to the usual caveats about historical simulation, is methodologically as defensible as the data permit.

The revisions made in response to instructor feedback further sharpen the methodology. The upgrade from plain OLS to a feature-engineered, ℓ_2-regularized Enhanced OLS specification is a strictly fairer baseline against the Random Forest, and the explicit in-sample vs. out-of-sample decomposition of the MVO results is a clean diagnostic of the "error-maximization" phenomenon that drives the gap between MVO and equal-weight performance.

From a results standpoint, the Random Forest-based strategy delivers a Sharpe ratio above 1 and a maximum drawdown below 15% over a 47-month out-of-sample period that spans both bear and bull regimes. The pipeline has therefore demonstrated that a methodologically simple smart-beta framework — 17 engineered factor and stock-level features for the linear branch, rolling Random Forest for the non-linear branch, six selected stocks, long-only Sharpe-maximizing MVO — can generate institutionally acceptable risk-adjusted returns when the model and the constraints are well calibrated to each other.

7.2 Weaknesses and Limitations

The methodology has four important weaknesses. First, the Enhanced OLS variant of the pipeline performs poorly enough that it would not be deployable, and the gap between Enhanced OLS+MVO (Sharpe 0.35) and RF+MVO (Sharpe 1.10) shows that the framework is critically dependent on the quality of the predictive signal. Even after the feature engineering recommended by the instructor's feedback, the linear-in-coefficients form keeps the cross-sectional spread of predictions tight enough that top-6 selection from the OLS branch is materially noisier than from the RF branch.

Second, the diagonal covariance approximation discards the information contained in pairwise stock correlations; in a more concentrated regime where a small number of factors drive most of the cross-sectional variation, this simplification could produce concentrated factor exposures without the optimizer's awareness. Third, the augmented feature set (FF5 + lagged returns + squared lags + factor–size interactions + log size), while substantially richer than a pure factor model, still omits firm-specific signals such as earnings surprises, analyst revisions, and news sentiment that are likely to carry incremental information at the monthly horizon. Fourth, the frictionless-trading assumption almost certainly overstates realized performance; transaction costs, liquidity constraints, and market-impact effects would all reduce the strategy's net Sharpe ratio in production, and the comparison between MVO and equal-weight strategies is particularly sensitive to turnover assumptions.

A fifth, more subtle concern — now made explicit by the in-sample vs. out-of-sample diagnostic — is that the equal-weight benchmark on the RF-selected stocks outperforms RF+MVO on a Sharpe basis out of sample, even though RF+MVO dominates equal weighting in sample by

construction. This is a reminder that any optimization layer must justify itself empirically, not just theoretically; a layer that systematically subtracts value out of sample should be replaced with a simpler default, or its inputs should be improved until the in-sample dominance survives the trip through estimation noise.

7.3 Future Work

Several extensions would meaningfully strengthen the framework. First, incorporating realistic transaction costs and turnover penalties directly into the optimization objective would penalize gratuitous reshuffling and produce weight vectors that are more stable across rebalances. Second, replacing the diagonal covariance matrix with a Ledoit–Wolf shrinkage estimator

$$\widehat{\boldsymbol{\Sigma}}_{\text{shrink}} \;=\; \delta\,\boldsymbol{F} + (1-\delta)\,\boldsymbol{S}, \tag{25}$$

where $\boldsymbol{S}$ is the sample covariance, $\boldsymbol{F}$ is a structured target (e.g., constant correlation), and $\delta \in [0,1]$ is the optimal shrinkage intensity, or with a factor-model covariance, both of which are known to outperform the sample estimator in out-of-sample portfolio applications and would allow the optimizer to capture pairwise correlations in a stable way. Third, enriching the feature set further — adding momentum at longer horizons, profitability quality, idiosyncratic-volatility signals, and analyst revisions — and re-running the comparison between Enhanced OLS and Random Forest on the richer feature set; we would expect the Random Forest's relative advantage to grow as the feature set widens, since the marginal benefit of capturing non-linear interactions increases with the number of features. Fourth, properly modelling the cross-sectional plus time-series (CSTS) panel structure flagged by the instructor with a true panel estimator (e.g., fixed-effects pooled OLS, Fama–MacBeth cross-sectional regressions, or a heterogeneous-coefficient Bayesian panel model) is the natural next step beyond the pooled-regression specification used here. Fifth, the gap between RF+MVO and RF equal-weight invites investigation of whether a less aggressive concentration cap (say, 1/4 instead of 1/3) or a switch to risk-parity weighting on the same selections would close that gap. Finally, a CVaR-based optimization variant would be informative once the holdings universe is enlarged enough to estimate the tail reliably.

7.4 Key Takeaways

The project demonstrates that a methodologically simple smart-beta pipeline — an augmented 17-feature design for the linear branch, rolling Random Forest for the non-linear branch, six selected stocks, long-only Sharpe-maximizing MVO with a diagonal covariance estimator — can deliver institutionally relevant out-of-sample performance when the predictive model is non-trivial. The Random Forest variant achieves a Sharpe ratio above 1 with a maximum drawdown below 15% over a 47-month backtest. The most important methodological lesson, made explicit by the new in-sample vs. out-of-sample diagnostic, is that selection quality dominates optimization quality on small holdings universes, and that the MVO layer's in-sample dominance over equal weighting can flip out of sample when predictions are noisy. The cleanest takeaway is that the combination of a flexible machine-learning model and a constrained mean–variance optimizer provides a transparent foundation on which more elaborate signals, more sophisticated risk models, and more realistic trading frictions can be layered as future research directions.

8. Appendices

Appendix A — Main References

- Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
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- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Kozak, S., Nagel, S., & Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2), 271–292.

Appendix B — Primary Data Sources

- CRSP Monthly Stock File, accessed via Wharton Research Data Services (WRDS): https://wrds-www.wharton.upenn.edu/
- Fama–French Five-Factor Data, Kenneth R. French Data Library: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Appendix C — GitHub Repository

All code, intermediate datasets, and the rendered methodology document are available in the project repository:

https://github.com/HaojingCheng-Jade/IDS586_Project_2026

The repository follows the structure recommended in the project overview:

- `data_process/Clean_2.ipynb` — data cleaning, merging, and feature-engineering notebook that produces the analysis-ready panel `data_clean_2.csv`.
- `Modeling/new_OLS.ipynb` — updated rolling-window Enhanced OLS estimation (`RidgeCV` with 17 engineered features, leave-one-out CV over a 13-point log-spaced grid for λ) and monthly top-6 selection (output: `data/new_ols_ridgecv_top6_by_month_post2022.csv`, 282 stock-month observations across 47 rebalance months).
- `Modeling/RandomForest.ipynb` — rolling-window Random Forest estimation and monthly top-6 selection (output: `data/rf_top6_by_month_post2022.csv`, 282 stock-month observations across 47 rebalance months).
- `MVO/new_MVO.ipynb` — updated long-only Sharpe-maximizing mean–variance optimization with diagonal covariance and per-name cap of 1/3, equal-weight benchmark construction, full backtesting, *and the new in-sample vs. out-of-sample tracking* described in Section 5.5 (output: `is_vs_oos_wealth_curves.png` and the IS/OOS performance tables).
- `measurement.ipynb` — post-hoc measurement notebook that loads the top-6 selection files for both estimation models and produces the cumulative-return comparison plots used in Section 6.

- `smart_beta_methodology.pdf` — formal mathematical methodology document accompanying the codebase.
- `README.md` — project overview, repository structure, data sources, methodology summary, run instructions, and team-member list.

The notebooks are intended to be run sequentially in the order listed; each stage depends on the outputs of the previous one. All intermediate data products are committed to the `data/` subdirectory of the repository so that the optimization and backtesting steps can be reproduced without re-running the upstream cleaning and modeling stages.